Lake Erie: where the NASA–local difference comes from

14 August 2024 • Hylak_id 9 • Sentinel-3B scene a59e9caf…

Same lake. Different counting units. Different QA and value encoding.

9,354 NASA-only detections: 5,153 QA exclusions + 4,201 spectral-stage nondetections.

2,221 local grid-cell detections represent 1,690 distinct native coordinates.

The slides separate measured discrepancies from causes that still need original scene/raster evidence.

The Erie CSV is linked to the supplied scene

The actual pairs file was found in Downloads; its counts match your first screenshot.

Evidence	Verified identity / role
ALL_INDEX_STATS	5,229 lake records × 50 columns; Erie is Hylak_id 9
Scene ID	a59e9caf-9ade-4115-affc-6a845651bd3d
Acquisition start	2024-08-14T15:37:11.380881Z • Sentinel-3B
Erie pairs CSV	286,772 NASA grid-cell rows × 23 columns
Second screenshot	Lake Superior example; its counts do not describe Erie

Same-scene filename is verified. Original NASA TIFF version and exact processing-run lineage remain unverified.

Match the record: scene and lake features

These 10 features identify the observation and lake; none is a cyanobacteria count.

Feature	Erie value
scene_id	a59e9caf-9ade-4115-affc-6a845651bd3d
scene_name	S3B OLCI EFR • 20240814T153711 → 154011[1]
acquisition_start	2024-08-14T15:37:11.380881Z
Hylak_id / Lake_name / Lake_area_km2	9 / Erie / 25,767.79 km²
state_name / state_id	Both missing in the source CSV
latitude / longitude	42.904334 / −78.908084 • lake metadata point

[1] Exact full product name is preserved in speaker notes and the feature mapping CSV. Lake metadata coordinates are not per-pixel coordinates.

Match the record: seven QA features

All numbers below count native observations from this scene.

Feature	Value	Meaning
native_lake_pixels	265,008	Stored native lake observations
primary_retained_pixels	233,050	Retained under primary QA
generic_flag_exclusion_sensitivity_pixels	231,727	Retained under strict QA
bloom_rescued_pixels	1,323	Flagged pixels kept by rescue rule
CLDICE_pixels	33,281	Cloud / ice flagged
cloud_excluded_pixels	31,958	Flagged and not rescued
bloom_rescue_fraction_of_CLDICE	0.03975241	1,323 / 33,281 = 3.9752%

265,008 = 233,050 + 31,958 • 33,281 = 31,958 + 1,323 • 233,050 = 231,727 + 1,323

The native CI funnel explains the 1,721 detections

Only 0.7385% of retained native pixels meet the final detection rule.

Feature / stage	Native pixels	What the next filter removes
ci_input_valid_pixels	233,050	226,056 fail the CI candidate test
ci_candidate_pixels = CI_valid_pixels	6,994	5,273 fail the final detection test
ci_cyano_detection_pixels = CI_cyano_valid_pixels	1,721	Final detected subset

SS(681) < 0 → CI = –SS(681)
Then SS(665) > 0 and CI > 0.0001

CI and CI_cyano summarize different subsets

A higher CI_cyano mean is expected after selecting cyanobacteria detections.

Statistic suffix	CI_<suffix>	CI_cyano_<suffix>
valid_pixels	6,994 candidates	1,721 detections
mean	0.00533793	0.01886181
median	0.00108173	0.01789697
p10	0.00013630	0.00278139
p90	0.02045253	0.03575557
max	0.06051304	0.06051304

Each row maps to the exact prefixed column name. p10/p90 are percentiles of finite values in that subset, not detection rates.

Other indices use all 233,050 retained pixels

Their means cannot be interpreted as means of the cyanobacteria-only subset.

Statistic suffix	NDCI_<suffix>	MPH_<suffix>	FAI_<suffix>
valid_pixels	233,050	233,050	233,050
mean	-0.10301671	0.00032487	-0.00111702
median	-0.10593817	0.00000767	-0.00130498
p10	-0.11874502	-0.00036254	-0.00280788
p90	-0.09045651	0.00047180	-0.00004025
max	0.40540335	0.12878148	0.11860112

A negative lake-wide NDCI or FAI mean can coexist with positive CI_cyano in a small bloom subset. These indices measure different spectral features.

Corresponding counts use different populations

Left: native observations. Right: NASA target rows selecting those observations.

Native summary feature	Native count	Corresponding target count
primary_retained_pixels	233,050	251,718
generic_flag_exclusion_sensitivity_pixels	231,727	250,013
bloom_rescued_pixels	1,323	1,705
CLDICE_pixels	33,281	36,744
ci_input_valid_pixels	233,050	251,718
ci_candidate_pixels	6,994	8,832
ci_cyano_detection_pixels	1,721	2,221

The right column sums pair flags, including repeated selections. Exact pairs-column counterparts are in slide notes and Erie_count_crosswalk.csv.

What the two spectral tests actually calculate

SS means spectral shape: reflectance minus a straight-line spectral baseline.

$$SS(681) = r681 - r665 - (r709 - r665) \times 16/44$$

$$SS(665) = r665 - r620 - (r681 - r620) \times 45/61$$

Negative SS(681) creates a CI candidate; CI = –SS(681).

Positive SS(665), sufficient CI and QA are also required for CI_cyano.

r is local rhos reflectance. Ratios use nominal wavelengths for explanation; code uses the actual wavelengths from the input product.

How one NASA cell selects a local pixel

Nearest native observation first; QA is evaluated after the spatial selection.

1. Take a NASA raster cell centre inside the selected lake polygon.

2. Search all finite native coordinates in the same lake and scene.

3. Choose the nearest point on the sphere; accept only within the distance limit.

4. Read that point’s QA and CI masks; require both detections for the paired plot.

The screenshot states 450 m; accepted distances reach 449.668 m. No farther clear-pixel substitute, footprint-overlap weighting, or one-to-one assignment.

The 2,221 versus 1,721 count is reconcilable

Grid rows repeat native coordinates; comparison to the summary assumes the same run.

Quantity	Count	Unit
Local detected in the native summary	1,721	Native observations
Distinct selected detection coordinates	1,690	Native coordinates in pairs
Repeat assignments beyond the first	531	Additional NASA target rows
Local detections in the pairs CSV	2,221	NASA target rows
Native-summary minus selected coordinates	31	Not represented, if exports are identical-run

1,690 + 531 = 2,221 • 1,721 − 31 + 531 = 2,221. The increase is sampling multiplicity, not 500 additional biological detections.

Recomputed detection table matches the screenshot

Detection flags alone do not distinguish valid nondetection from missing data.

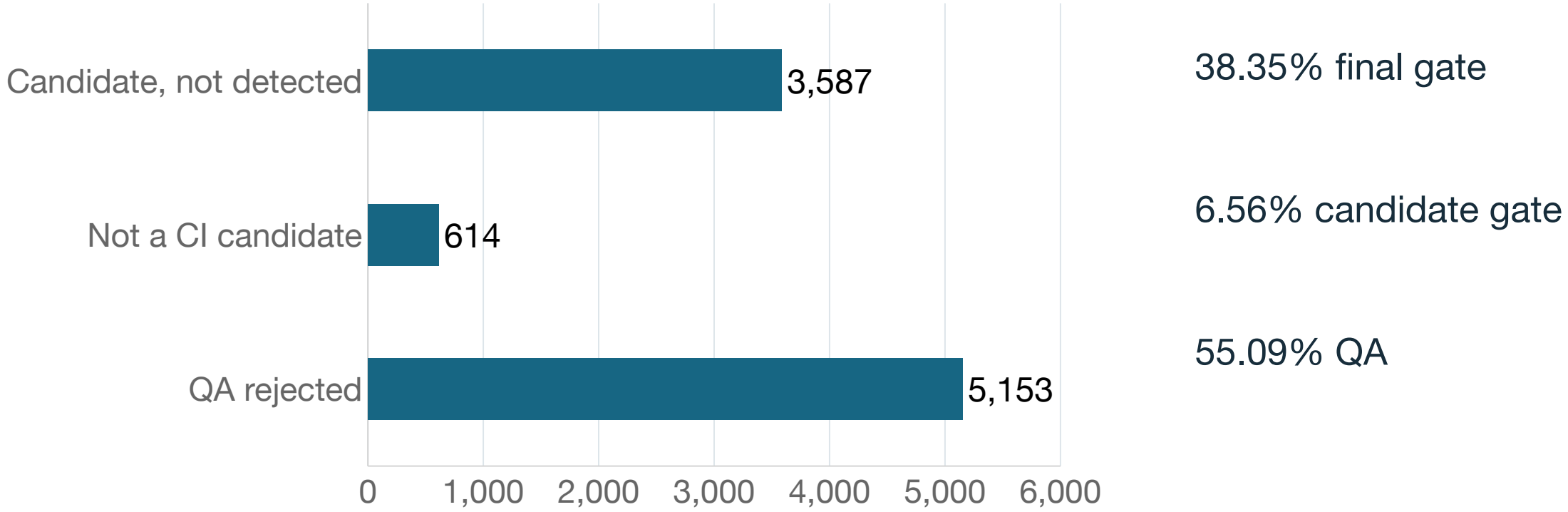
NASA flag	Local detected	Local not detected	Row total
Detected	2,183	9,354	11,537
Not detected	38	275,197	275,235
Column total	2,221	284,551	286,772

18.92% of NASA-detected rows are also local detections.

98.29% of local-detected rows are also NASA detections.

Exactly where the 9,354 NASA-only rows stop

The categories below are mutually exclusive and cover every NASA-only row.



Zero fail matching; zero fail finite CI input after QA. The final-gate group cannot be split into SS(665) vs CI threshold without spectral inputs.

Bloom rescue strongly affects the paired sample

The paired set depends on your QA policy.

Pairs-CSV measure	Primary QA	Strict QA
QA-valid target rows	251,718	250,013
Local detection target rows	2,221	516
Both detected target rows	2,183	513

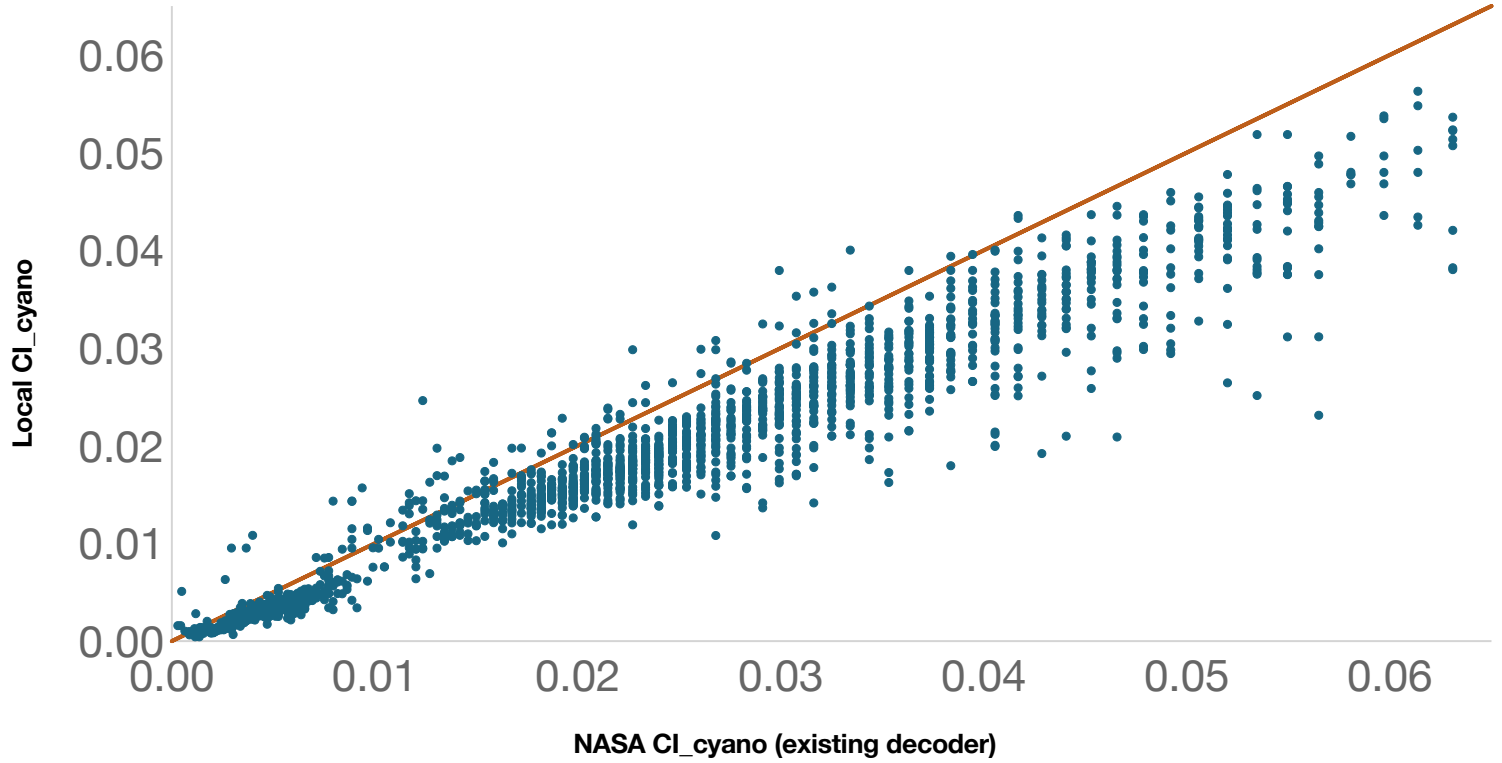
1,670 of 2,183 paired rows (76.50%) select bloom-rescued pixels.

All 5,153 QA-rejected NASA-only rows select unrescued CLDICE pixels.

“Rescued” means retained by a spectral rule. It is not independent proof that the cloud flag was wrong.

High correlation coexists with a value offset

2,183 paired NASA grid rows; both CI_cyano values exceed 0.0001.



r = 0.9690

NASA mean 0.02315
Local mean 0.01846

Bias = –0.00469
(local – NASA)

Orange line is 1:1. High correlation shows covariation; it does not demonstrate equal values or agreement on bloom extent.

DN decoding changes the magnitude comparison

Sensitivity test on the same 2,183 rows isolates the decoding effect.

Metric	Existing decoder	V6 formula scenario
NASA mean	0.02314950	0.02061362
Local mean	0.01845999	0.01845999
Bias: local – NASA	–0.00468952	–0.00215363
MAE	0.00490739	0.00274560
RMSE	0.00629129	0.00381251
Pearson r	0.969012	0.969230

RMSE falls 39.4%; about 54.1% of the mean offset is removed. This supports a decoder contribution, conditional on the TIFF being Version 6.

The decoder also changes detection membership

Existing code: 10^(0.012 × DN – 4.2), accepting DN 0–250.

NASA V6: 10^(0.011714 × DN – 4.1870866)

Issue	Observed consequence in this Erie file
V6 DN 1–253 are data	15 local detections at DN 251–253 were excluded
V6 DN 0 / 254 / 255	Below threshold / land / no data
DN 18 crosses CI > 0.0001	Five additional NASA-only detections in the V6 scenario
V6 scenario totals	11,557 NASA detections; 2,198 paired; 23 local-only

NASA-only becomes 9,359. Verify the TIFF processing version before changing the production decoder. These are recomputed scenarios, not a rerun of NASA processing.

A “cell” can mean three different things

Separate spatial counts, index values, and estimated biological cell density.

Term	Meaning	Erie example
Detected pixels / grid cells	Number of spatial observations	1,721 native; 2,221 target rows
CI_cyano	Spectral index magnitude	Native detected mean = 0.01886181
Estimated cells/mL	Empirical CI_cyano × 100,000,000	Illustrative mean ≈ 1,886,181 cells/mL

The CSV stores CI values and pixel counts, not measured organism counts.

The illustrative mean applies only to the detected native subset—not the whole lake.

A worked match: nearby pixels, different values

Source CSV row 117,875 (header counted as row 1).

Field	Value
NASA centre (latitude, longitude)	42.15222020, –80.10680701
Selected native centre	42.15089035, –80.10701752
local_match_distance_m	148.8878 m
obpg_raw_dn → obpg_CI_cyano	172 → 0.00731139 (existing decoder)
local_CI_cyano	0.00577824
Final result	Both detected; local – NASA = –0.00153315

Distance verification across all 286,757 matched rows agrees within 2.7×10^{-9} m. This verifies the exported distances, not nearestness against an unavailable full native archive.

Check concrete rows at each failure stage

These rows demonstrate why a spatial match is not a detection match.

Original CSV row	NASA DN	Distance	Local stopping point
38,623	25	174.33 m	CLDICE; selected QA fails
39,237	134	165.53 m	QA passes; not a CI candidate
42,069	19	100.06 m	Candidate; final detection fails
117,874	254	133.07 m	Local detected; NASA classified as land

Full coordinates, flags and CI values are in Erie_worked_examples.csv. A missing local CI value is not automatically a zero concentration.

The 38 local-only rows are not one failure type

NASA DN categories explain the exported flag disagreement.

NASA DN	Rows	Interpretation under V6
0	6	Below NASA detection threshold
251 / 252 / 253	3 / 6 / 6	Data rejected by the existing local decoder
254	9	NASA land mask differs from local lake support
255	8	NASA no-data classification
Total	38	35 of these rows select bloom-rescued pixels

DN 255 does not identify the exact missing-data cause. Do not label all such rows “cloud,” or all local-only detections “false positives.”

What is proven—and what is still unresolved

Different products can disagree without either one being a ground-truth measurement.

Factor	Evidence status
Many-to-one spatial assignment	Measured: 2,221 target detections → 1,690 coordinates
Local QA and spectral gates	Measured: 5,153 + 614 + 3,587 NASA-only rows
DN conversion and range	Measured sensitivity; source TIFF version unverified
Observation timing and mapping	NASA daily merged maximum / area-weighted grid; local single scene
Upstream spectral processing	Possible residual cause; matched reflectances and run metadata needed

The first two rows explain count semantics and filter outcomes. They do not prove why the underlying reflectances or cloud tests disagree.

How to interpret the correlation fairly

The 2,183 paired rows reuse only 1,662 native coordinates.

The paired plot excludes 9,354 NASA-only rows and 38 local-only rows.

Repeated native selections weight some observations more than others.

The paired mean (0.01845999) differs from the native detected mean (0.01886181) because the support and weights differ.

For validation, report coverage, detection disagreement and value error separately.

Corrections to carry into your diagrams

The previous comparison method is useful; lake-specific findings must be recomputed.

Replace the Lake Superior example in image 2 with the verified Erie funnel.

“Before selected QA” already includes upstream primary QA; it is not an unfiltered spectral detection.

“Strict” checks selected flags—not every possible quality flag.

Use “retained by bloom rule,” rather than claiming cloud misclassification was proven.

Pairs features: NASA and spatial matching

One pairs row is a NASA target cell, not a unique native observation.

Exact feature name(s)	Interpretation
latitude / longitude	NASA cell centre
obpg_raw_dn	Encoded NASA raster value
obpg_science_valid	Existing decoder validity, not an independent QA truth
obpg_CI_cyano	CI decoded from DN
obpg_detected_gt_0_0001	NASA valid and CI > 0.0001
local_match_found / local_match_distance_m	Accepted nearest match / spherical distance
nearest_native_latitude / nearest_native_longitude	Coordinates of selected native observation

Pairs features: QA and local CI stages

False flags on an unmatched row are defaults, not observed negative measurements.

Exact feature name(s)	Interpretation
nearest_CLDICE / nearest_HISATZEN / nearest_NAVFAIL	Stored flags at accepted match
nearest_bloom_rescued	Retained through bloom rescue rule
nearest_generic_strict_valid	Local strict QA selector
nearest_primary_bloom_aware_valid	Local primary QA selector
nearest_selected_QA_valid	Selector chosen for comparison
local_ci_input_valid / local_ci_candidate	Finite QA input / candidate
local_ci_detection_before_selected_QA	Stored primary-QA detection
local_ci_cyano_detected / local_CI_cyano	Final detection / stored CI
paired_above_0_0001_for_correlation	Both sides valid and above threshold

The next test should isolate one cause at a time

Think about the observation you want to compare before altering the algorithm.

First: verify the NASA TIFF version, daily product and original scene/run metadata.

Then: recover SS(665), SS(681), CI and flags for the 3,587 candidate failures.

Next: compare matching temporal support and spatial support; retain a unique-native sensitivity analysis.

Question to discuss: do you want agreement with NASA’s daily product, or equivalence for one overpass?

Evidence and reproducible handoff

The deck preserves exact feature names; values are rounded only for display.

Erie_feature_value_mapping.csv — all 50 features in original column order.

Erie_pairs_audited.csv — every original row and column, plus reason and CSV row number.

Erie_audit_summary.json — counts, metrics, source paths and SHA-256 hashes.

Erie_worked_examples.csv — inspect actual flags and coordinates behind the explanations.

Scientific references and code locations are in slide notes. No production code was changed and no NASA/local reprocessing was performed.